BOARDGAME AFICIONADOS

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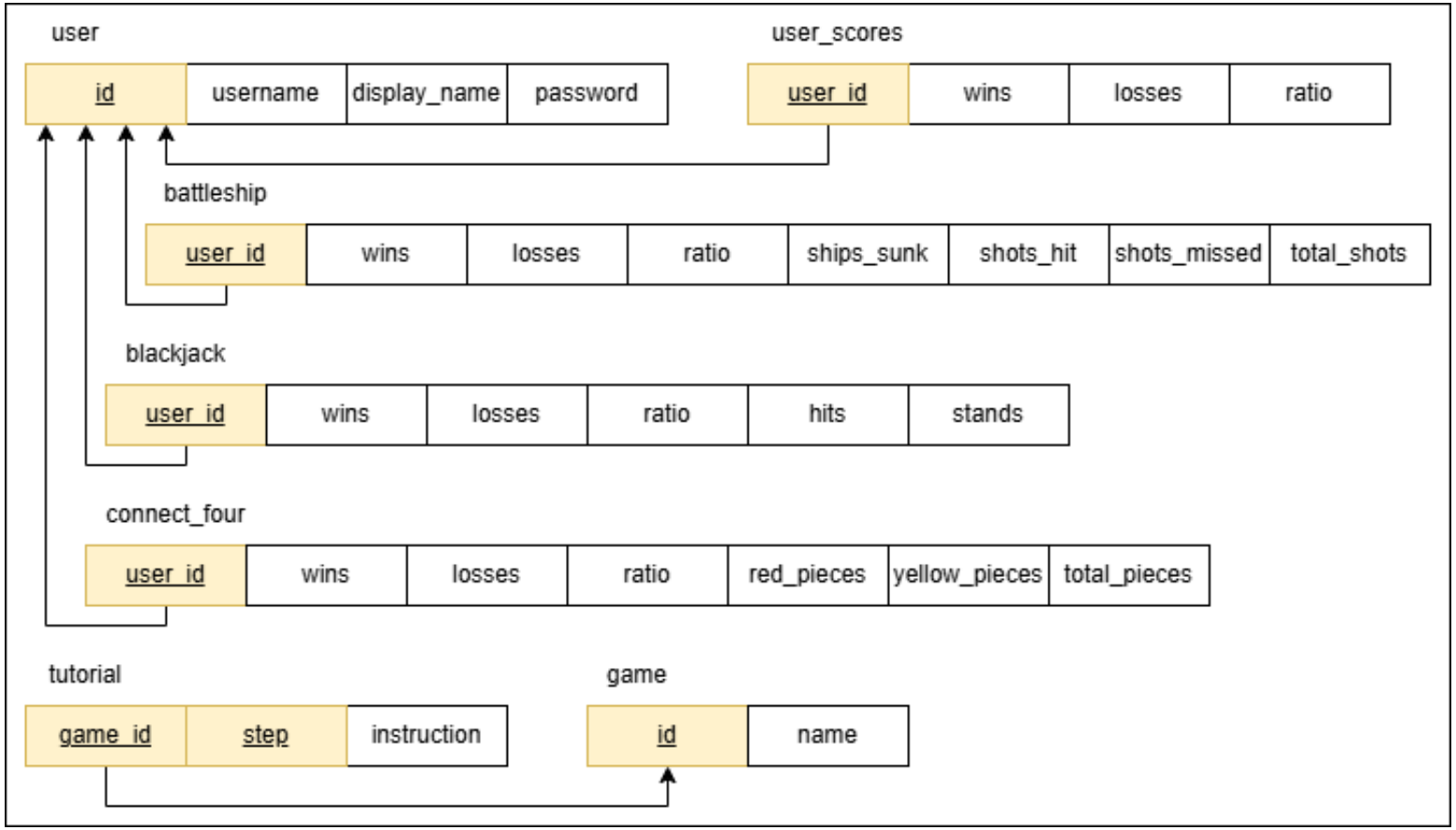
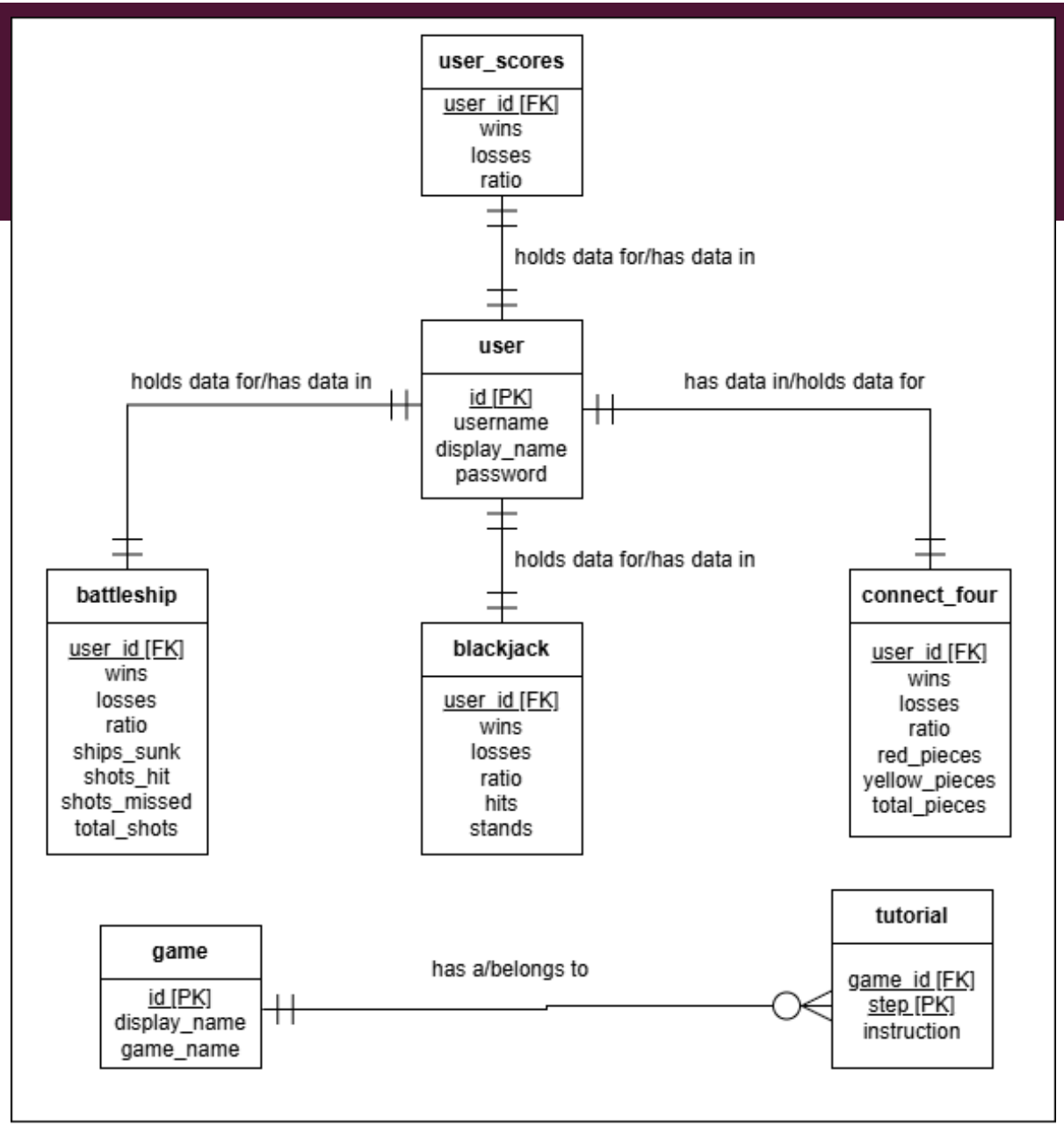
The Boardgame Aficionados have the goal to up the ante on boardgames for anyone and everyone. Making use of a web application approach to boardgames allows easy play of anything that exists, and live stats tracking for pitting yourself against your family and friends!

SQL DATABASE SCHEMA

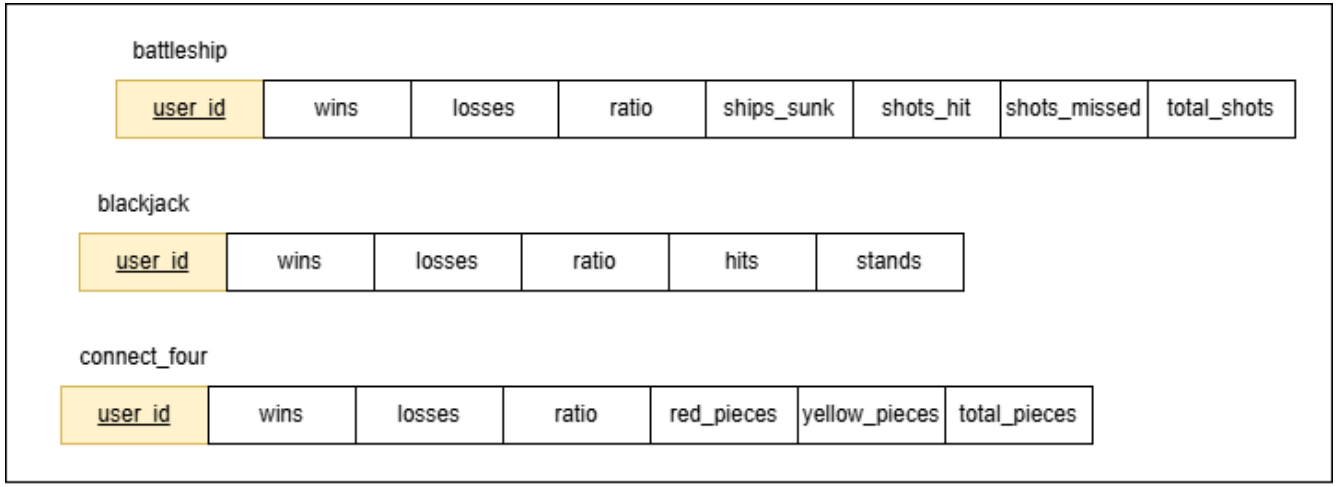


ER DIAGRAM


user_scores
user_id [FK]
wins
losses
ratio
holds data for/has data in
user
id [PK]
username
display_name
password
holds data for/has data in
has data in/holds data for
holds data for/has data in
battleship
user_id [FK]
wins
losses
ratio
ships_sunk
shots_hit
shots_missed
total_shots
blackjack
user_id [FK]
wins
losses
ratio
hits
stands
connect_four
user_id [FK]
wins
losses
ratio
red_pieces
yellow_pieces
total_pieces
game
id [PK]
display_name
game_name
has a/belongs to
tutorial
game_id [FK]
step [PK]
instruction

GAMES

Each game is unique from the others. The stats they hold in common are user_id, wins, losses, and ratio. However, these cannot be merged into a separate table, as they are unique per game, per user. Each row in these tables is the statistics of the user specified in user_id.



user_id	wins	losses	ratio	ships_sunk	shots_hit	shots_missed	total_shots
1	6	4	1.5	14	42	33	75
2	4	6	0.67	10	35	41	76
3	4	3	1.33	18	52	41	93
4	8	4	2	34	96	70	166
5	1	10	0.1	5	21	88	109
6	2	0	2	10	28	12	40
7	0	0	0	0	0	0	0
8	5	7	0.71	22	61	73	134
9	12	2	6	51	140	60	200
10	5	0	5	25	70	20	90

Battleship is shown above

MY TOP 4 SQL QUERIES

```sql
-- 5. Who is the best player per game? SELECT (SELECT u.display_name FROM user u WHERE u.id IN ( SELECT ba2.user_id FROM battleship ba2 WHERE ba2.ratio IN ( SELECT MAX(ba1.ratio) FROM battleship ba1 ) )) AS 'Best Battleship Player', (SELECT u.display_name FROM user u WHERE u.id IN ( SELECT bl2.user_id FROM blackjack bl2 WHERE bl2.ratio IN ( SELECT MAX(bl1.ratio) FROM blackjack bl1 ) )) AS 'Best Blackjack Player', (SELECT u.display_name FROM user u WHERE u.id IN ( SELECT c2.user_id FROM connect_four c2 WHERE c2.ratio IN ( SELECT MAX(c1.ratio) FROM connect_four c1 ) )) AS 'Best Connect 4 Player';
```

[Edit inline] [Edit] [Create PHP code]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

Best Battleship Player	Best Blackjack Player	Best Connect 4 Player
Phil	Logan	Mike

```sql
-- 7. Based on total matches, what game is played the most? SELECT game AS 'Most Played Game', MAX(total_games) AS 'Total Games' FROM ( SELECT 'Battleship' AS game, SUM(ba.wins + ba.losses) AS total_games FROM battleship ba UNION SELECT 'Blackjack' AS game, SUM(bl.wins + bl.losses) AS total_games FROM blackjack bl UNION SELECT 'Connect 4', SUM(c.wins + c.losses) AS total_games FROM connect_four c ) high_plays GROUP BY game ORDER BY MAX(total_games) DESC LIMIT 1;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Most Played Game	Total Games
Blackjack	87

MY TOP 4 SQL QUERIES (CONT.)

```
-- 3. What are the average wins per game? SELECT (SELECT AVG(ba.wins) FROM battleship ba) AS 'Battleship Average Wins', (SELECT AVG(bl.wins) FROM blackjack bl) AS 'Blackjack Average Wins', (SELECT AVG(c.wins) FROM connect_four c) AS 'Connect 4 Average Wins';
```

[Edit inline] [Edit] [Create PHP code]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

Battleship Average Wins	Blackjack Average Wins	Connect 4 Average Wins
4.7000	3.7000	3.6000

```
-- 8. What players have never lost in at least one game? SELECT u.display_name AS '0 Losses' FROM user u WHERE u.id IN ( SELECT ba.user_id FROM battleship ba WHERE ba.losses = 0 UNION SELECT bl.user_id FROM blackjack bl WHERE bl.losses = 0 UNION SELECT c.user_id FROM connect_four c WHERE c.losses = 0 );
```

[Edit inline] [Edit] [Create PHP code]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

0 Losses
Mike
Nikki
Phil
Sheldon

PROBLEMS WITH SQL

1.

Many games have expansion packs, and multiple variants available. Because of this, there could be additional statistics to add later that rely on that expansion pack. The trait of a table not including a game_id due to the way the data is means that adding additional tables is not an option for these, as you could not correlate it back to the original.

2.

Statistics in future games could be much more suited for arrays or dictionaries. A game like Monopoly being added, could track what properties you have owned, and what level they have reached. This would be an unreasonable number of columns in SQL, meaning a dictionary is the most viable option.

3.

A lot of the purpose of this database is to put people on a leaderboard. The more games are added, the more difficult this would become due to the complexity of games varying. The three shown (Battleship, Blackjack, Connect 4) are all simple and hold purely numerical data. Games like Monopoly, Clue, Catan, etc. would have a wide range of data types present, as well as far more present. This would be very difficult to base queries on.

SOLUTIONS WITH MONGODB

1.

MongoDB is a solution here as you can easily add more data entry points. You don't need to be worried about breaking the schema and adding more columns since adding columns is as easy as that. Its ability to decide what data is shown in a query also prevents the problem of having far too much data be shown just for one game entry per player.

2.

MongoDB supports dictionaries unlike SQL. This would be easy to fix, and it could all be under one attribute (property_levels in the Monopoly example) to boot.

3.

MongoDB has more data-type support, which would already reduce the complexity of the data stored. However, it has more advanced operations that can be performed as a result as well, giving you a much easier means to lay out and properly compare the data for the sake of a leaderboard.

MONGODB COLLECTIONS

- boardgame-aficionados
  - battleship
  - blackjack
  - connect_four
  - games
  - users

users and user_scores were merged
game and tutorial were merged

```
_id: ObjectId('69e69
user_id : 1
wins : 6
losses : 4
ratio : 1.5
ships_sunk : 14
shots_hit : 42
shots_missed : 33
total_shots : 75
```

battleship above

```
_id: ObjectId(
user_id : 1
wins : 7
losses : 5
ratio : 1.4
hits : 29
stands : 18
```

blackjack above

```
_id: ObjectId('69e69
user_id : 1
wins : 5
losses : 3
ratio : 1.67
red_pieces : 58
yellow_pieces : 53
total_pieces : 111
```

connect_four above

```
_id: ObjectId('69e69990c99915017029dedf')
id : 1
game_name : "battleship"
display_name : "Battleship"
tutorial : Array (5)
  0: "Each player places ships of fixed len
  1: "Players alternate calling a coordinat
  2: "Opponent responds “hit,” “miss,” or “
  3: "Mark results and continue taking turn
  4: "First player to sink all enemy ships
```

games above

```
_id: ObjectId('69e6995cc99
id : 1
username : "vjbrazier"
display_name : "Vincent"
password : "***"
total_wins : 18
total_losses : 12
total_ratio : 1.5
```

users above

TOP 2 MONGODB QUERIES

What user is in the middle of the overall rankings?

```
db['users'].find(
  {},
  {
    '_id': 0,
    'display_name': 1
  }
)
.sort({
  'total_ratio': 1
})
.skip(4)
.limit(1)
{
  display_name: 'Taylor'
}
```

Who has more than 10 overall wins, but a <1 overall ratio?

```
db['users'].find(
  {
    $and: [
    {'total_wins': {$gt: 10}},
    {'total_ratio': {$lt: 1}}
    ]
  },
  {
    '_id': 0,
    'display_name': 1
  }
)
{
  display_name: 'Taylor'
}
```